5 Coherent light is incident normally on a double slit, as shown in Fig. 5.1.

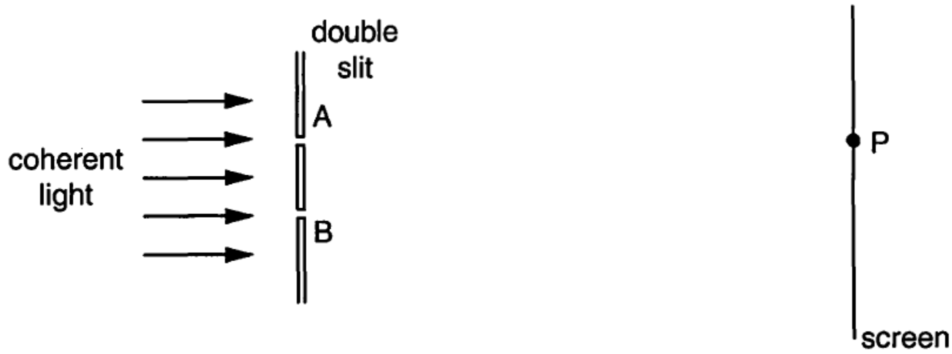


Fig. 5.1 (not to scale)

Light passes through the two slits A and B and is incident on a screen.

The variation with time *t* of the displacement *x* of the light arriving at point P on the screen is shown in Fig. 5.2.

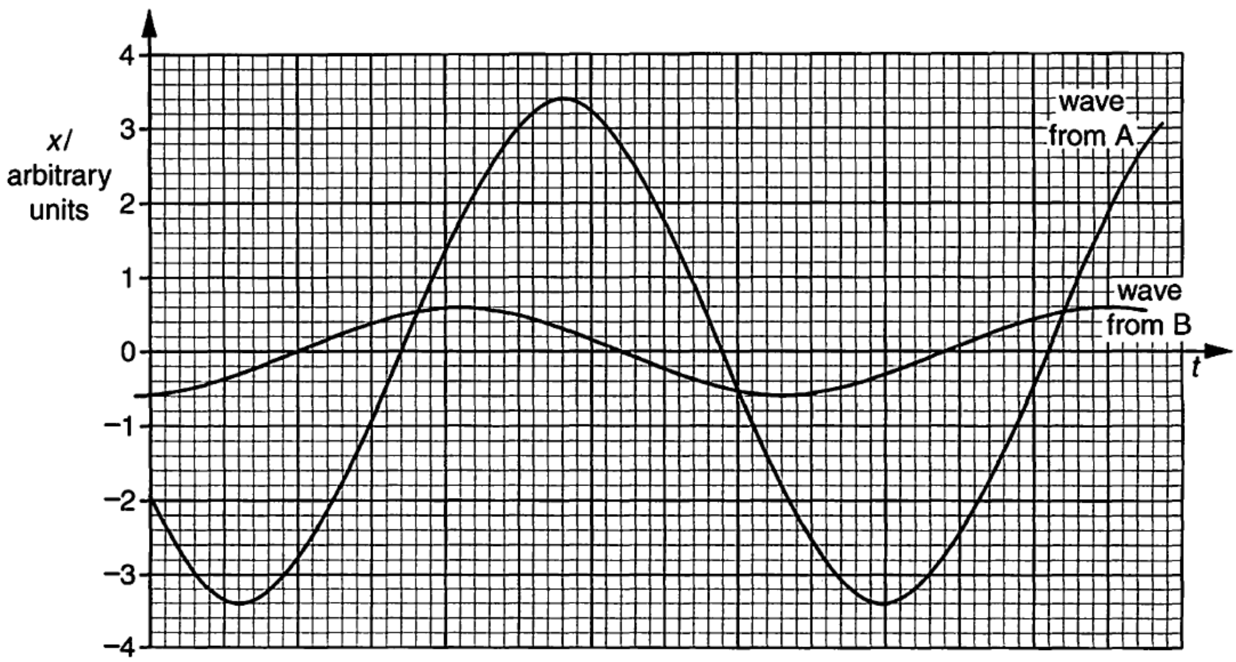
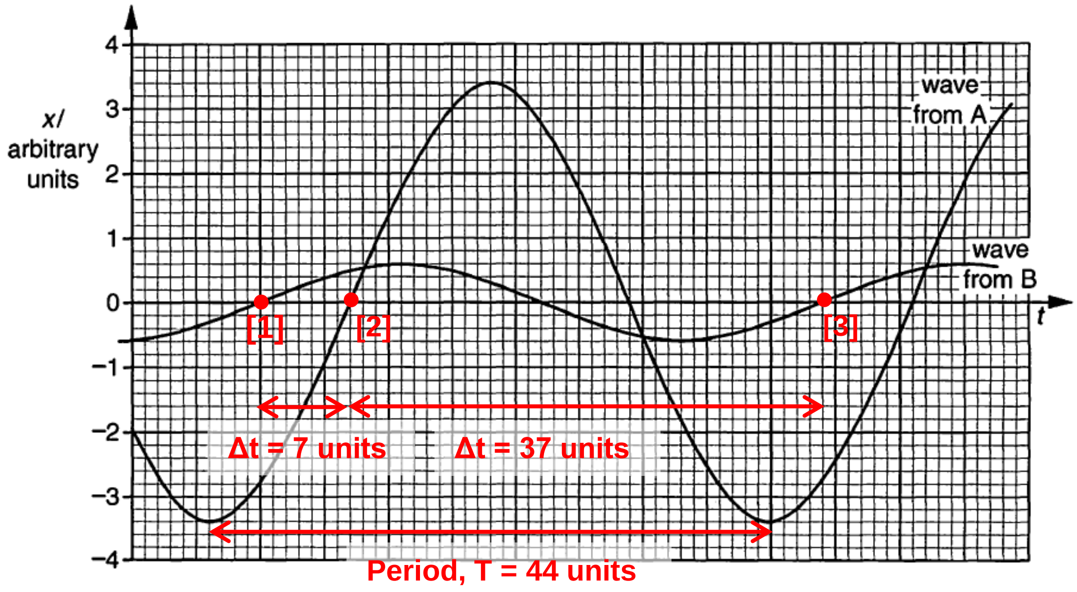


Fig. 5.2

(a) Use Fig. 5.2 to determine the phase difference between the waves from slit A and from slit B that arrive at point P.

phase difference = ...° [2]

L2 Determine the time difference Δt from between points [1] and [2], or, between [2] and [3].



$\frac{\Delta\varphi}{360^\circ} = \frac{\Delta t}{T}$ **M1**

$\frac{\Delta\varphi}{360^\circ} = \frac{37}{44}$ **A1**

$\Delta\varphi = \frac{37}{44}x360^\circ = 302.73^\circ$

Phase difference = **303° (or 300°).**

OR

$\frac{\Delta\varphi}{360^\circ} = \frac{7}{44}$

$\Delta\varphi = \frac{7}{44}x360^\circ = 57.3^\circ$

Phase difference = **57.3° (or 57°)**

(b) Dark fringes and bright fringes are both formed on the screen.

Use Fig. 5.2 to determine, for the bright fringe and the dark fringe closest to point P, the ratio

$$\frac{\text{intensity of light at the bright fringe}}{\text{intensity of light at the dark fringe}}.$$

ratio = ... [3]

L2 At a **bright** fringe, the waves arrive **in phase**, undergoing **constructive interference**. Thus the resultant amplitude
= 3.4 + 0.6 = 4.0 units

At a **dark** fringe, the waves arrive **in antiphase (180˚ out of phase)**, undergoing **destructive interference**. Thus the resultant amplitude
= 3.4 – 0.6 = 2.8 units **C1**

		(Since we are considering the dark and bright fringes **closest** to P, assume that the amplitudes of the waves from A and B do not differ significantly from those arriving at P) *[Recall that when slits are not infinitely small, single slit diffraction effects causes the double slit bright fringes to be non-uniform in intensity. If the bright fringes are close, the difference in intensity is less.]* The intensity of a wave I is directly proportional to the square of its amplitude A. $\frac{I_{bright}}{I_{dark}} = \left(\frac{A_{bright}}{A_{dark}}\right)^2 = \left(\frac{4.0}{2.8}\right)^2$ $\frac{I_{bright}}{I_{dark}} = 2.0408 = 2.0$	**M1** **A1**
	(c)	In an attempt to produce brighter fringes, the student widens each of the two slits, keeping their separation constant. Fringes are no longer observed. Suggest why the fringes are no longer observed.	
		....	
		....	
		....	
		....	[2]
	L3	When the two slits are widened, the **light waves passing through each slit will diffract less.** A smaller degree of diffraction causes the region in which the two waves overlap and interfere to become smaller. **If the widths of the slits are too wide, the two waves do not overlap at all and hence no interference occurs**. Hence, the fringes are no longer observed. ***Note: As a coherent light beam is used (e.g. laser beam), widening the slits do not cause the waves to be incoherent. Citing incoherence as the reason is not acceptable.***	**B1** **B1**

[Total: 7]

6	Two metal plates X and Y are contained in an evacuated container and are connected as shown